Analyzing the Impact of Socioeconomic Variables on Education

Data Science and Analytics

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Introduction & Data Overview

Numerous studies have shown that among high school students, communities or regions with a ***higher socioeconomic status*** also show ***higher academic performance*** (Bradley). Often, schools in communities of low socioeconomic status receive less funding and resources to provide for students, which may negatively affect their academic progress. Performing well in high school is crucial as it plays a key role in students' admissions into college and realization of scholarship opportunities, later on playing a role in the success of their careers and levels of income. As a result, limited access to opportunities during high school hinders students' ability to create strong foundations for their future careers.

Data was collected from ***credible***, ***open-access sources*** to conduct an exploratory analysis of the relationship between socioeconomic variables and educational outcomes. After consolidating the data, we focused on the following variables to provide greater insight into the issue at hand.

- Household income
- Health insurance coverage
- Government spending on education
- State-by-state political alignment
- Total school enrollment
- ACT scores
- Advanced course enrollment
- Student-to-teacher ratio

Data Dictionary

Our primary dataset (*Children-Enrolled Public: Economic Characteristics*) utilizes a system of codes for column names.

Code	***Type***	***Data***
CDP03_2P	Percent	Total households with all parents in the labor force (Children under 6 years)
CDP03_4P	Percent	Total households with all parents in the labor force (Children over 6 years)
CDP03_5E	Number	Total Income and Benefits
CDP03_6P	Percent	Total households w/ less than $10,000 yearly income
CDP03_7P	Percent	Total households w/ $10,000-$14,999 yearly income
CDP03_8P	Percent	Total households w/ $15,000-$24,999 yearly income
CDP03_9P	Percent	Total households w/ $25,000-$34,999 yearly income
CDP03_10P	Percent	Total households w/ $35,000-$49,999 yearly income
CDP03_11P	Percent	Total households w/ $50,000-$74,999 yearly income
CDP03_12P	Percent	Total households w/ $75,000-$99,999 yearly income
CDP03_13P	Percent	Total households w/ $100,000-$149,999 yearly income
CDP03_14P	Percent	Total households w/ $150,000-$199,999 yearly income
CDP03_15P	Percent	Total households w/ more than $200,000 yearly income
CDP03_16E	Number	Median household income (dollars)
CDP03_23P	Percent	Total households with Earnings
CDP03_25P	Percent	Total households with Social Security Income

CDP03_27P	Percent	Total households with Retirement Income
CDP03_29P	Percent	Total households with Supplemental Security Income
CDP03_31P	Percent	Total households with Cash Public Assistance
CDP03_33P	Percent	Total households with Food Stamp/SNAP Benefits
CDP03_49P	Percent	With private health insurance coverage
CDP03_50P	Percent	With public health insurance coverage
CDP03_51P	Percent	With no health insurance coverage
CDP03_54P	Percent	Families whose income is below the poverty level
STATEFP	Number	State FP ID
STATEABBREV	String	State 2 letter abbreviation (ex. TX)

Purpose

This analysis seeks to identify various means by which ***socioeconomic status may impact high school students***. Although it is known that a certain portion of students in the United States grow up under impoverished circumstances, because this group represents only a small portion of the population, they are often overlooked (Wai et al.).

To analyze the effects of socioeconomic status on these students, these variables were tested against educational outcomes to see how they relate. The conclusions drawn from this analysis can assist lawmakers in producing ***legislation*** that takes into account the plight of various groups of students. Furthermore, they can help lead the United States toward a state in which socioeconomic status has a minimal effect on students' educational outcomes, ensuring ***equality of opportunity*** for all students. The results of this analysis can help inform decisions regarding:

- Educational policy
    - Government spending (Yearly Spending per Student)
    - Course offerings (Advanced Math, Science, Technology courses. CTE education and pathways)
    - Extracurricular funding (Summer programs, CTE competitions)
    - Hiring of teachers
- Taxation policy
    - Property taxes (Effect on Spending per Student)
    - Income & Payroll taxes (Effect on financial burden on families)
- Healthcare policy
    - Federal insurance programs (Medicare, Medicaid, ACA)
    - Medical debt policy

Methods

Our methods are comprised of six (6) techniques:

1. **Obtaining and cleaning data from *trusted* sources with free, open access**:
   Data was converted into the form of comma-separated value spreadsheets (.csv). The data was formatted into simple rows and columns. Entries with empty values were eliminated. Our data sources were
   a. *National Center of Education Statistics* (ACS-ED 2014-2018 Children-Enrolled Public: Economic Characteristics (CDP03)
   b. *The U.S. Census Bureau* (Public Education Spending by State)
   c. *The U.S. Department of Education* (Civil Rights Data Collection Office for Civil Rights 2017-18 State and National Tables)
   d. *ACT* (State-by-State Average Test Scores)
   e. *FiveThirtyEight* (State-by-State Partisan Lean Scores)
2. **Summary Statistics**:
   We made extensive use of the **mean**, **median**, and **standard deviations** of various columns in our datasets. These statistics were plotted in various ways through the use of **maps, histograms**, **scatter plots**, and **pie charts**.
3. **Local Regression**:
   LOWESS regression was used to model non-linear relationships between two variables by creating a line of fit with a constantly changing slope.
4. **K-Means Clustering**:
   This algorithm was used to ***group states*** in our analysis based on socioeconomic, educational, and geographic variables.
5. **Cosine Similarity**:
   This technique uses the ***dot product*** to find similarity and confirm correlations.
6. **Principal Component Analysis**:
   This is a ***dimension-reduction*** method that can determine which variables are most influential in determining educational outcomes.

We made heavy use of the programming language ***Python*** alongside the libraries **Pandas** (Manipulating data), **Plotly** (Creating graphs), and **Numpy** (Mathematics)

Results

Poverty in the United States

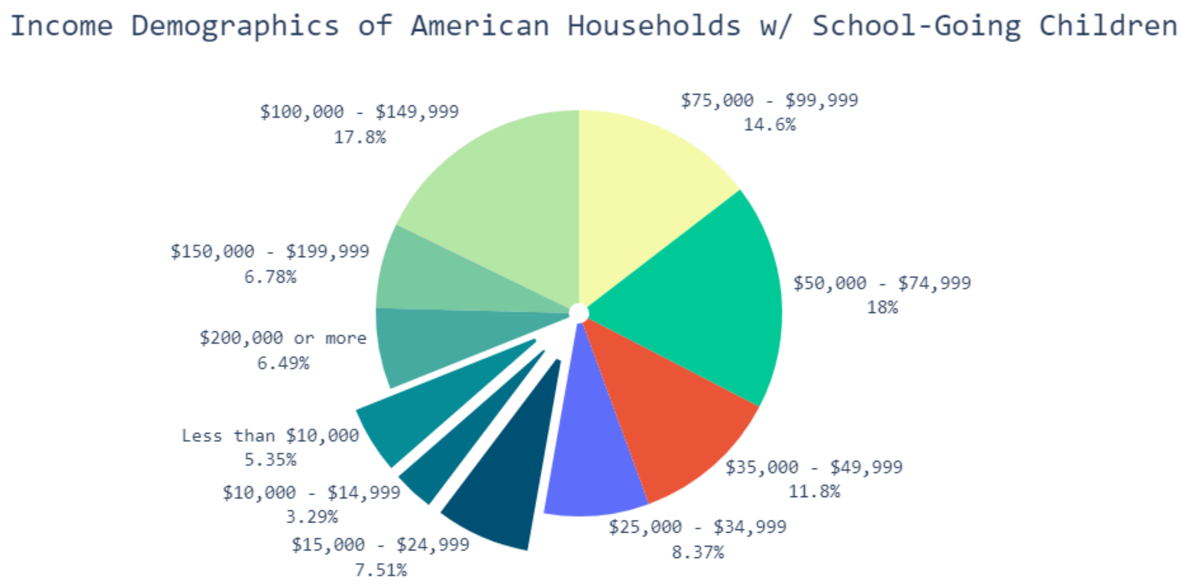


Figure 1. Pie chart showing the distribution of incomes among American Households with School-Going Children.

Poverty is a circumstance that is prevalent across the entire nation. In 2018, the year this data was collected, the poverty line for a single person was considered to be ***$12,140/yr***. However, this dataset is composed of families with at least one child. Thus, most of these families include at least three or four individuals. According to the U.S. Department of Health and Human Services, the poverty line for a family of four in 2018 was ***$25,100/yr***. Thus, a total of ***16.15%*** of the families in the dataset are considered to be below the poverty line according to these standards. This is equivalent to nearly ***1 in every 6*** families in the United States. The pie chart also shows that a proportion of ***31.07%*** of households have a yearly income of ***$100,000+/yr***. This demonstrates that, since the number of households below the poverty line is a relatively small percentage, this group may be overlooked.

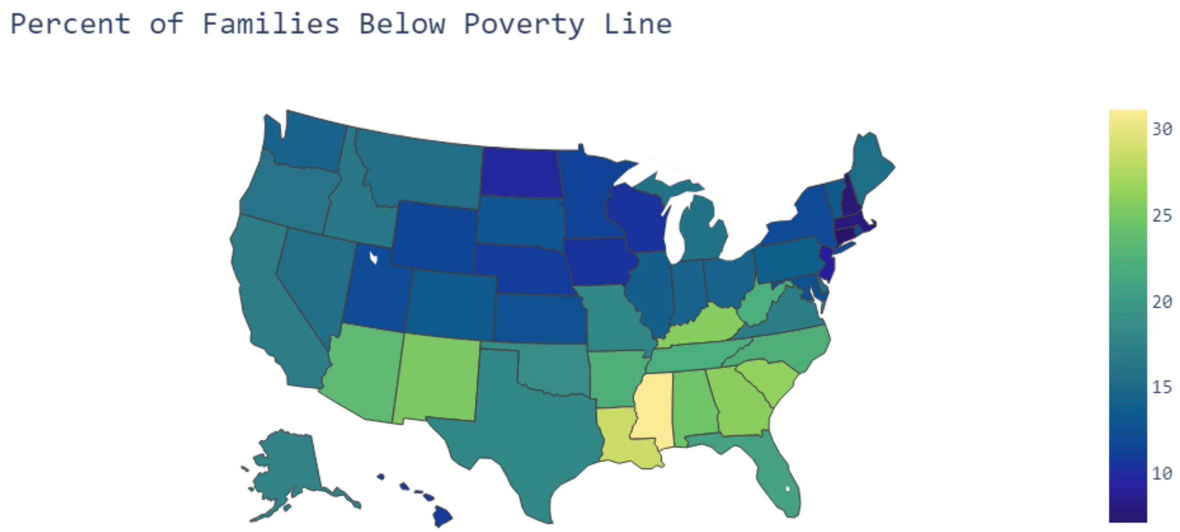


Figure 2. Map showing the state-by-state distribution of poverty rates across the nation.

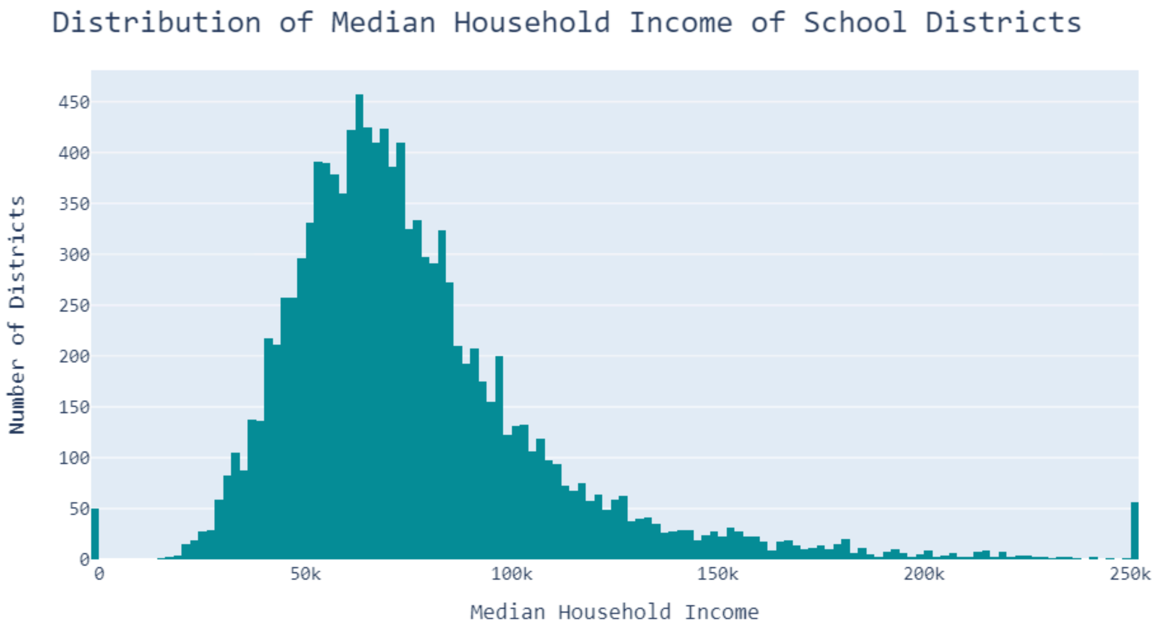


Figure 3. Histogram showing the distribution of Median Household Income across all School Districts.

To analyze the issue of poverty at a smaller scale of analysis, the percentage of households that are below the poverty line for each state was displayed on a nationwide map. The dataset provided median household income and the percentage of households below the poverty line at the school district scale of analysis. To construct this map, the districts were grouped by state and the ***mean*** was taken. This resulted in a ***standard deviation of ±9.44%***.

It can be seen that the proportion of households below the poverty line in Southern states is much higher than that which are Northern. Specifically, the states of Mississippi, Louisiana, Alabama, Georgia, and Kentucky show the ***highest rates*** of poverty. In contrast, the New England states of Connecticut, Massachusetts, New Hampshire, and New Jersey show the ***lowest rates*** of poverty. The Midwestern states also show somewhat low rates of poverty, while the states on the West Coast show higher rates of poverty, but still lower than the South.

Concerning median household income, it can be seen in Figure 3 that the data follows a shape similar to that of a ***normal distribution*** which is somewhat skewed to the right (peak left of center). The mean of the data is ***$76,471***, with quite a high standard deviation of ***±$33,579***. There are also some notable outlier districts (about 100) that show extreme values of median household income and do not fit within the distribution. A ***Kolmogorov-Smirnov test*** was utilized to test the data for normality, which further reinforces the hypothesis that the data adheres to a normal distribution.

Other important economic variables that can indicate the presence of economic hardship within a school district are the reliance of families on welfare programs such as Social Security, Retirement Income, Supplemental Security Income, Cash Public Assistance, and Food Stamp and SNAP Benefits. The number of families dependent on these services generally represents a small proportion of the total population (mean of ***10.53%***), but this can still play a significant role in educational outcomes. These factors are further investigated through the use of Principal Component Analysis.

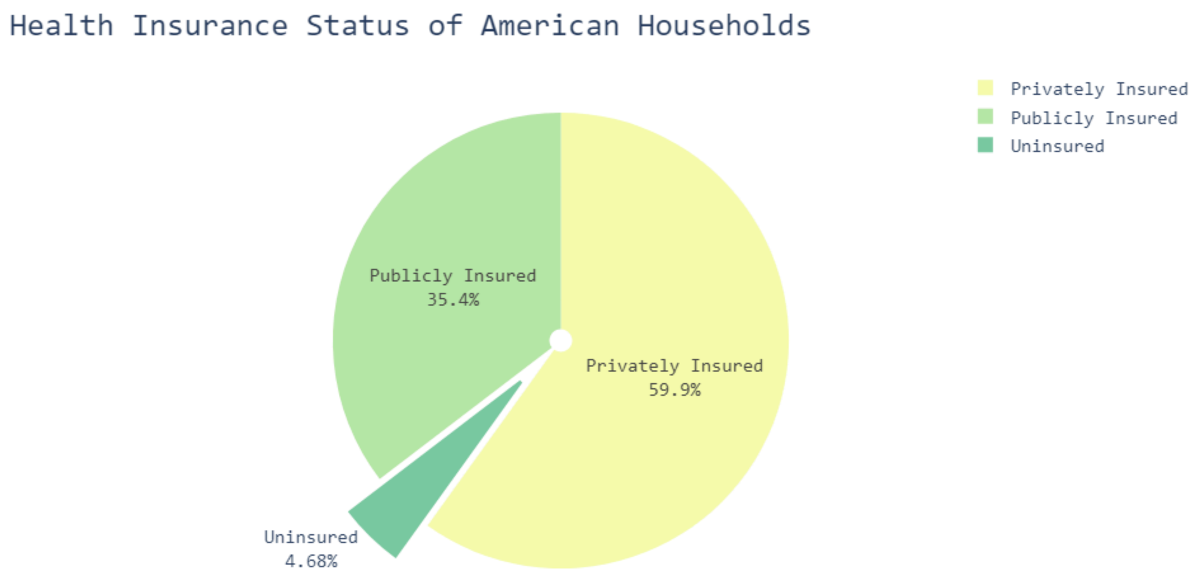


Figure 4. Pie chart showing the distribution of insurance status among American households with school-going children.

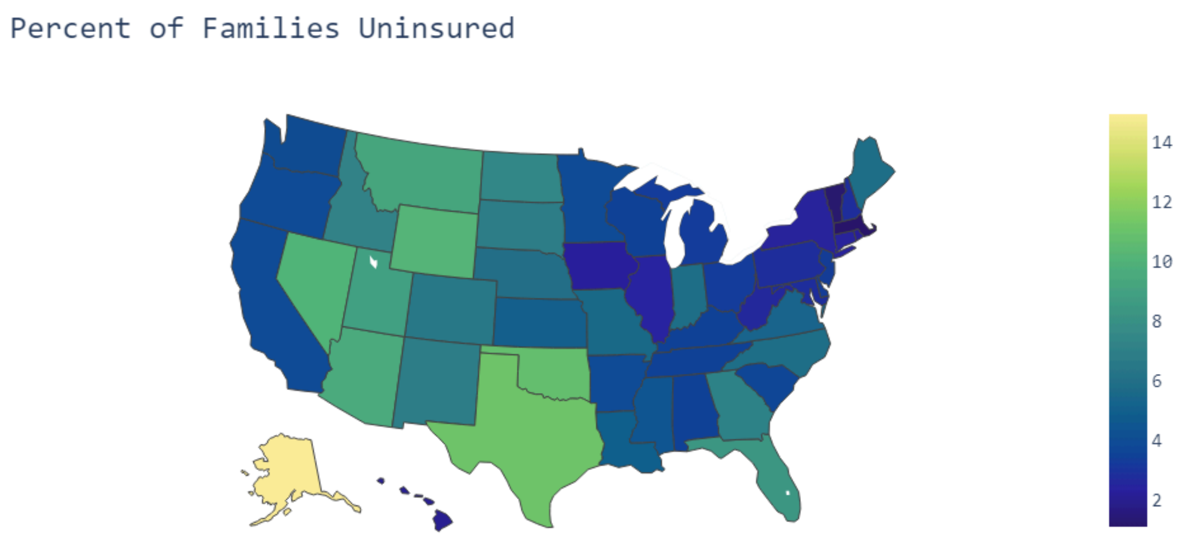


Figure 5. Map showing the state-by-state distribution of the percentage of households without health insurance coverage in the nation.

Health Insurance is one of the major factors other than income that determines a family's level of economic stability. To display the level of access to health insurance in the nation, a pie chart for the national scale of analysis was made, and a map showing the state-by-state distribution of uninsured households was made. Figure 4's results should be taken with a grain of salt since the standard deviations for the percentages of privately insured, publicly insured, and uninsured are quite high (***±19.01%, ±17.99%, and ±5.45%*** respectively). Figure 5 has a standard deviation of ***±4.31%***.

Figure 4 shows that approximately ***35.4%*** of households are dependent on some publicly funded insurance such as Obamacare, Medicare, or Medicaid. ***4.68%*** of households have no health insurance at all. Though health insurance is not an absolute necessity, it can be a point of great stress for families, for if there is some unforeseen medical emergency, it can mean going into medical debt for hundreds of thousands of dollars in some cases. While some portion of publicly insured households are under Medicare after retirement, many of these households are insured under programs that provide coverage for those who are not provided with private insurance by their employer or are not able to afford it.

Figure 5 shows that there is a geographic trend for health insurance coverage as well, but it is very different from the trend shown for household income. It can be seen that states east of the Great Plains region are well insured, some of which have close to ***0%*** of households uninsured. However, states to the West, except those bordering the Pacific (California, Oregon, Washington), seem to show higher rates of households being uninsured. Alaska shows an extremely anomalous proportion without insurance of ***14%***.

From these visualizations, it is clear that poverty is a ***prevalent*** issue among American households, but is present at a low enough rate to possibly be ***overlooked*** when analyzing its effects on education.

Government Spending on Education by State

Government Spending is known to be a crucial factor in the success of students. The extent to which spending influences educational outcomes will be evaluated in subsequent analyses, but it is important to first understand how government spending on education varies across the nation.

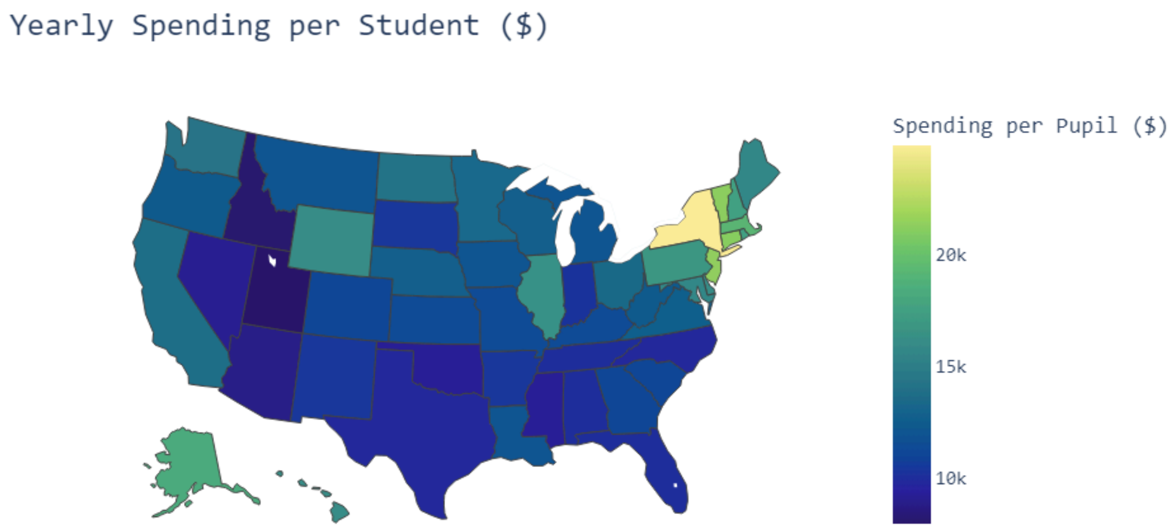


Figure 6. Map showing the state-by-state distribution of yearly government spending per student across the nation.

Figure 6 does not show a very strong geographic trend for government spending. However, it can be seen that Idaho and Utah specifically have extremely low spending per student (***below $10,000***), and New York and many of the other New England states have quite high levels of spending (***above $20,000***).

To draw a meaningful conclusion on government spending, a scatter plot graphing government spending by state against each state's political lean according to FiveThirtyEight's analysis after the 2020 election was graphed.

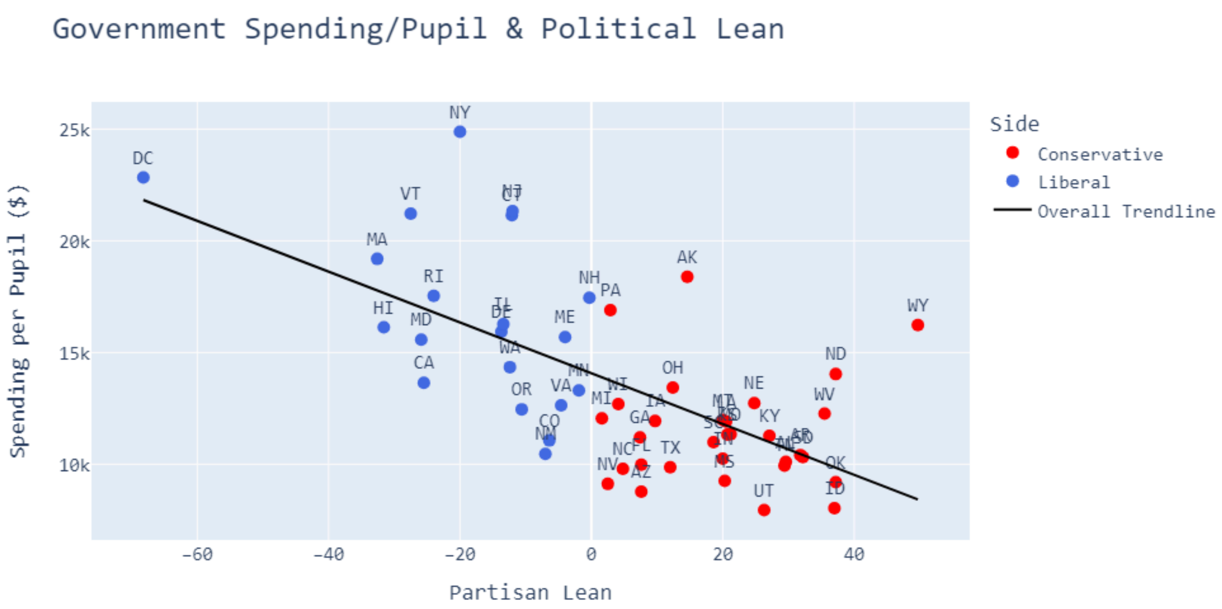


Figure 7. Scatterplot showing spending per pupil by state and partisan lean by state.

Figure 7 shows the spending per student on the vertical axis and partisan lean on the horizontal axis. A negative partisan lean indicates that the state is more liberal (left-leaning), while a positive partisan lean indicates that the state is more conservative (right-leaning). Though the association is relatively weak, there is a negative correlation between partisan lean and spending per pupil.

Least-squares linear regression was used to model this relationship. The creation of the trendline resulted in a correlation coefficient of ***$r = -0.648$*** (correlation of determination of ***$r^2 = 0.42$***). Although this value is not particularly strong, it does verify that there is a negative correlation between these variables.

This result concurs with the standings of the major political parties. The Republican party typically advocates for smaller government with lower taxes and less government spending, while the Democratic party advocates for larger government with higher taxes and greater government spending.

Effect on ACT Scores

The effectiveness of standardized test scores in determining student achievement has been debated over the past several years. Some argue that standardized testing does not allow students to express creativity and only shows how good students are at taking tests. However, standardized tests do provide a means of evaluating students objectively, rather than evaluating them on more abstract qualities which, while important, can be interpreted quite subjectively (Procon.org).

Unlike the SAT, the ACT tests a larger variety of subjects. While the SAT assesses English, reading, and writing in a combined manner and math separately, the ACTs assess reading, English, math, and science separately and produce composite scores. Therefore, due to the ***ACT's wider array of tested subjects***, the test was chosen to maintain a controlled analysis.

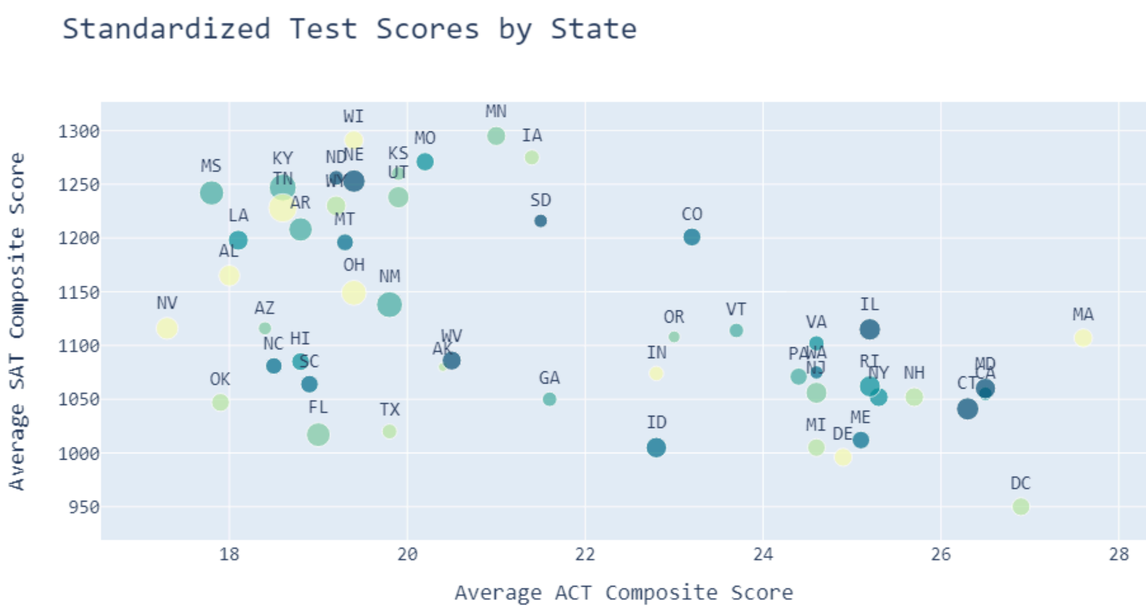


Figure 8. Scatterplot showing state-by-state average SAT and ACT scores.

Although the ACT was chosen, the SAT shows a very different state-by-state distribution of scores. Further research is needed to determine why these tests show ***drastically different*** outcomes.

First, we chose to analyze the relationship between ACT Scores and each state's spending per student.

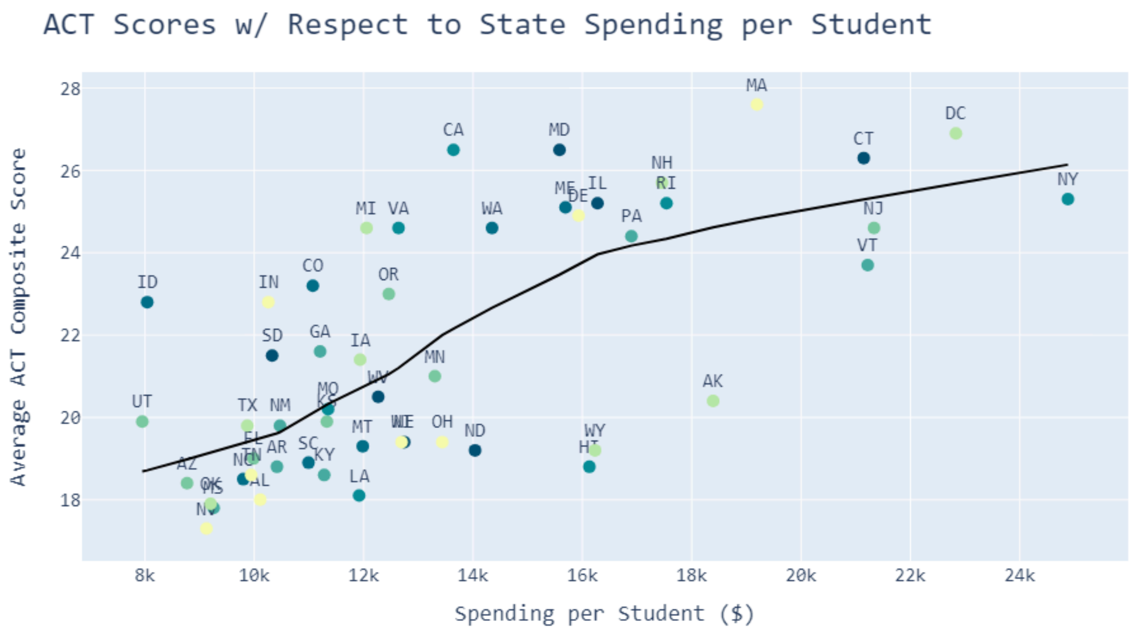


Figure 9. Scatterplot showing average ACT Composite scores against state spending per student w/ LOWESS regression line.

Figure 9 shows a ***weakly associated, positive correlation*** between ACT scores and spending per student. A significant proportion of the states analyzed showed ***low spending*** alongside a ***low average ACT score***. The scores then seem to take a relatively steep rise and then plateau. These states that show high spending per student and high average ACT scores seem to mainly fall within the New England region. It should be noted that there are arguably little to no states that have high spending per student and also low ACT scores, except for Alaska.

Locally weighted scatterplot smoothing, or ***LOWESS*** regression (a type of local regression) proved to be useful in this case as it was able to display a trend that was not exactly linear. Instead, the data in Figure 9 took the shape of an arc, which is not easily modeled by a straight line. The use of this model instead of classical least squares regression allows us to acknowledge the presence of a correlation even if it is not fully linear. The regression was done by breaking the points on the scatter plot into small chunks and finding the slope for each segment while ensuring that the intercepts at the ends of the interval match those of the subsequent and preceding intervals (Figueira). This allowed for the creation of a ***smooth curve*** with a changing slope.

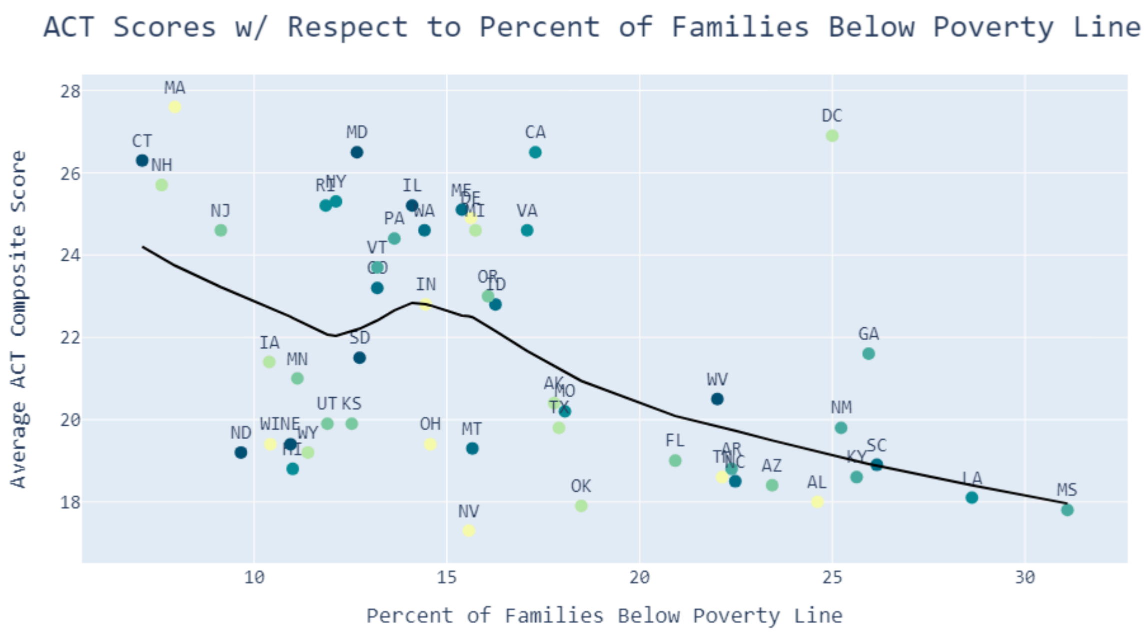


Figure 10. Scatterplot showing average ACT Composite scores against the percentage of families below the poverty line.

Although Figure 10 has a very different ***LOWESS*** curve, it tells a similar story to that told by Figure 9. In the region of states on the scatterplot with a low percentage below the poverty line, the average ACT composite scores tend to fluctuate and seem relatively random till ***17%***. However, above 17%, it can be seen that all of the states have relatively low average ACT scores, except for Washington DC. On the left side of the scatter plot, the distribution of states seems to be ***random***, but on the right side, there is a clear ***negative correlation*** between the two variables.

The two scatter plots show that household income and government spending on education have a clear influence on performance on the ACT. ***Government spending*** typically results in ***higher ACT scores*** and ***higher poverty rates*** result in ***lower ACT scores***.

Advanced Course Enrollment

Enrollment in advanced courses is another measure that can be used to evaluate educational outcomes. In contrast to standardized tests, advanced enrollment can show students' ability to handle intellectually challenging course content. However, one downside to this measure is that very specific STEM courses such as computer science, engineering, or AP sciences are not offered at many schools. Advanced Math classes can come in the form of introductory calculus, or even linear algebra or multivariate calculus in some schools. Thus, due to its ***diversity***, enrollment in advanced math courses was chosen as the statistic for measurement in this portion of the analysis.

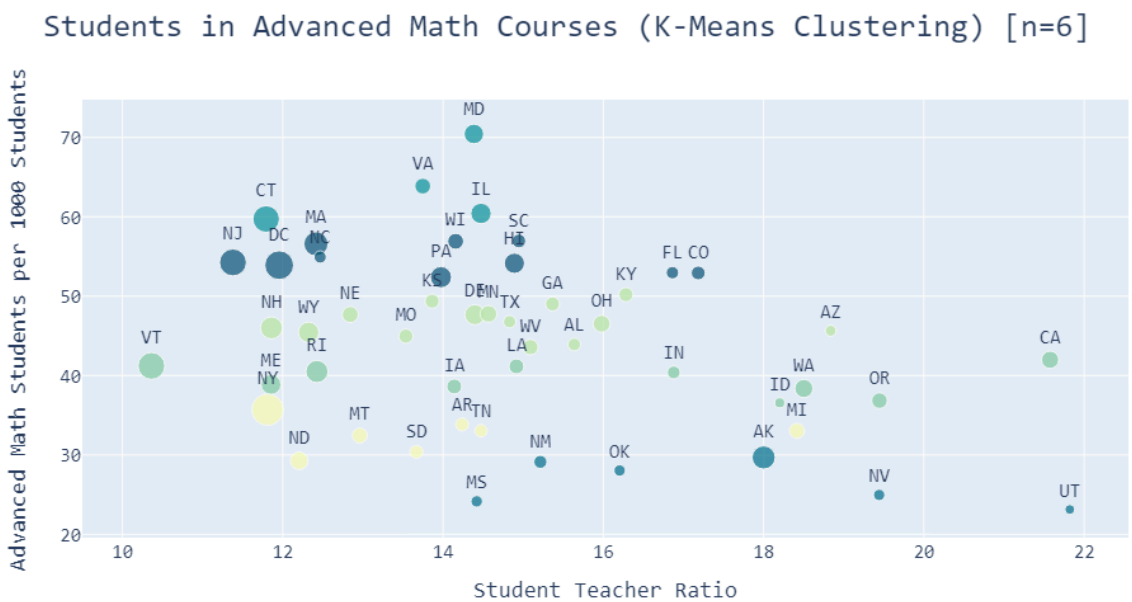


Figure 11. Scatterplot displaying clustered states based on student-to-teacher ratio [horizontal axis], advanced math enrollment [vertical axis], government spending per student [size of markers], and assigned cluster [color of marker].

One of the significant trends displayed in Figure 11 is that, for the most part, as the ***student-to-teacher ratios increase***, the size of the markers, representing each state's ***spending per student, decreases***. This shows that when there is more educational funding, there are more teachers to accommodate the students. However, there are not any other observable correlations between the variables shown. Instead, ***K-Means***

Clustering was used to group these states based on their student-to-teacher ratios and the enrollment of students in advanced math classes.

K-means clustering is done by calculating the ***distance*** between each data point and one of the k ***centroids*** *(Sharma)*.

$$d(x,\ y)\ = \sqrt{\sum_{i=1}^{n}(x_i - y_i)^2}$$

Figure 12. Distance Formula.

Then, the centroids are reinitialized.

$$C_i = \frac{1}{|N_i|}\sum x_i$$

Figure 13. Reassignment of centroids.

To determine the number of clusters, the ***elbow method*** was used. The point at which the graph has "***leveled off***" is said to be the ***optimal*** number of clusters. Six clusters were chosen as the optimal amount.

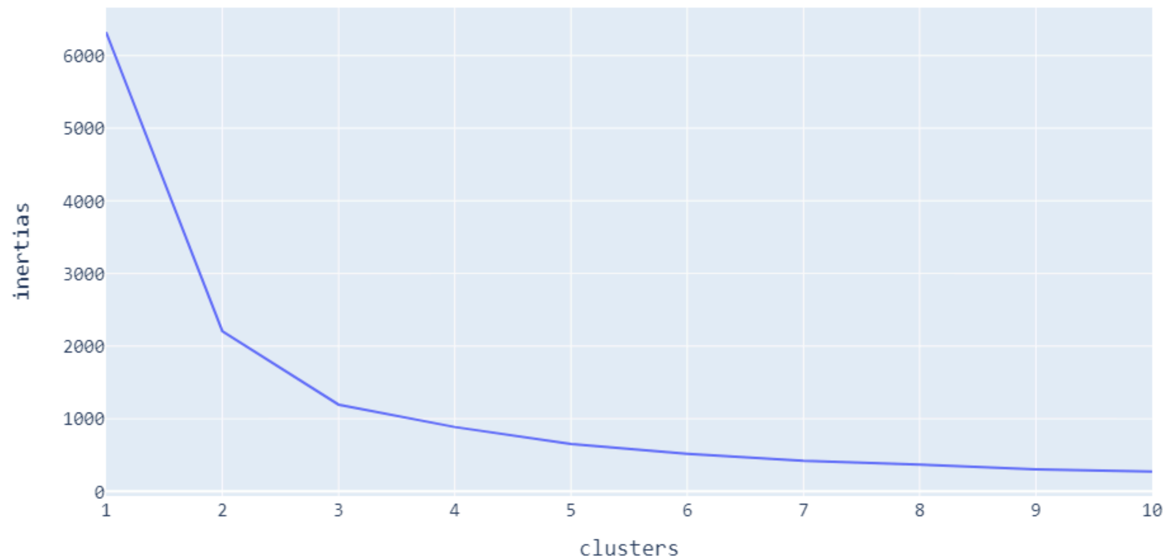


Figure 14. K-Means elbow method showing inertia values against the number of clusters.

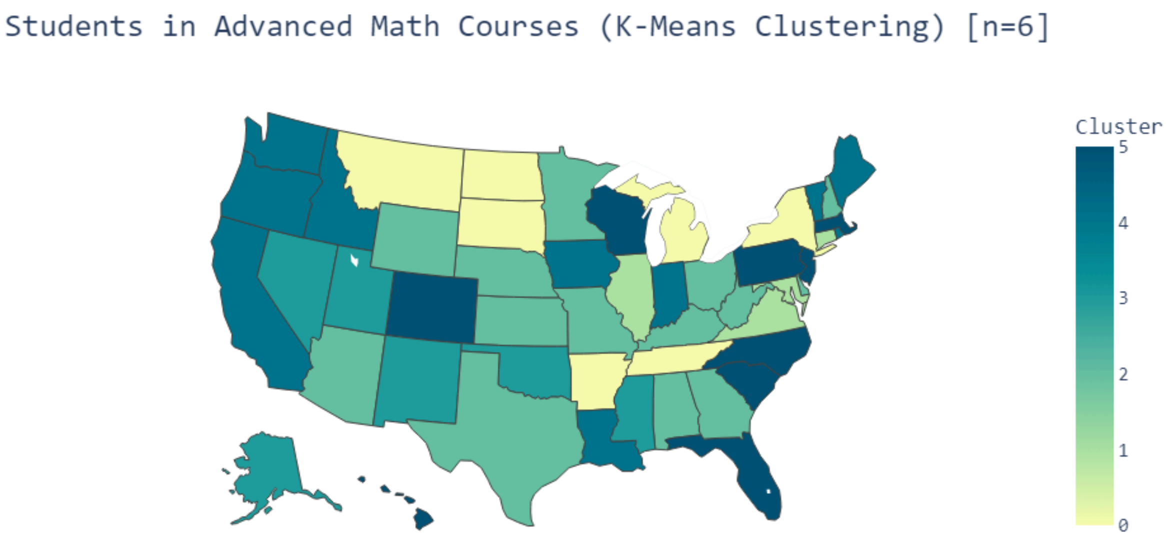


Figure 15. Map showing the cluster identification of each state based on its level of enrollment in advanced mathematics and its average student-to-teacher ratio.

Each of the six clusters has different profiles:

- **Cluster 0** contains states that show ***low*** student-to-teacher ratios and ***extremely low*** enrollment of advanced math students.
- **Cluster 1** contains states that show ***moderate*** student-to-teacher ratios and ***extremely high*** enrollment of advanced math students.
- **Cluster 2** contains states that show ***a variety*** of different student-to-teacher ratios and ***moderate*** enrollment of advanced math students.
- **Cluster 3** contains states that show ***somewhat high*** student-to-teacher ratios and ***extremely low*** enrollment of advanced math students.
- **Cluster 4** contains primarily ***Pacific coastal*** states that show ***very high*** student-to-teacher ratios and ***relatively low*** enrollment of advanced math students.
- **Cluster 5** contains states that show ***relatively low*** student-to-teacher ratios and ***relatively high*** enrollment in advanced mathematics.

Confirmation of Correlations

Cosine similarity is a frequently used measure of ***commonality***. The technique's premise is that the smaller the angle between two vectors of the same magnitude, the closer the values of each vector's components are to each other (Kniaz).

The cosine similarity value of two vectors is determined by taking the ***dot product*** of the two vectors, and dividing it by the ***product of the magnitudes*** of the two vectors:

$$cos(\theta) = \frac{\vec{A} \cdot \vec{B}}{||\vec{A}|| \, ||\vec{B}||}$$

Figure 16. Cosine Similarity Equation.

The dot product is found by taking the sum of the products of each of the components of the vectors. The magnitude of each vector is found by taking the sum of each of the squared components of the vectors and taking the square root.

$$cos(\theta) = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2} \sqrt{\sum_{i=1}^{n} B_i^2}}$$

Figure 17. Cosine Similarity Equation with Substitutions.

This was used to find the cosine similarity values for pairs of 6 different variables in our data. The representative vectors were created by taking the state-by-state values and applying a ***normalization*** to ensure magnitude does not skew the results.

$$\vec{v}_{normalized_i} = \frac{\vec{v}_i - min(\vec{v})}{max(\vec{v}) - min(\vec{v})}$$

Figure 18. Vector Normalization.

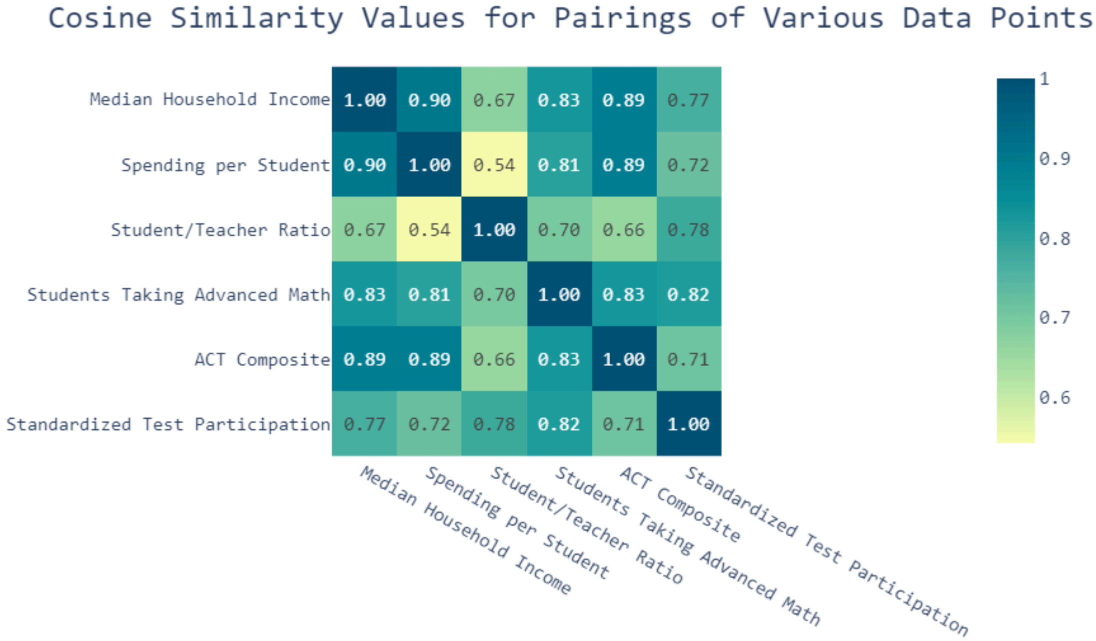


Figure 19. Cosine Similarity Values for Pairings of Various Data Points.

Figure 19 shows the calculated cosine similarity values. The ***closer*** a value is to 1, the ***more similar*** the two vectors are. The ***further*** a value is from 1, the ***more dissimilar***, or possibly negatively correlated the variables are. The diagonal in the center of the heatmap contains only values of 1.00. This is because the variables being compared across the diagonal are identical, so they are perfectly similar.

Notable values include:

- ***Median Household Income*** & ***Spending per Student***: Very strong similarity value of ***0.90***. This means that states that have high median household income typically have very high spending per student as well.
- ***ACT Composite*** & ***Spending per Student***: Strong value of ***0.89***. This confirms the analysis presented in Figure 8.
- ***Students Taking Advanced Math*** & ***Median Household Income***: Relatively high value of ***0.83***. This shows that though the association may not be very strong, a positive correlation is present.
- ***Spending per Student*** & ***Student-to-Teacher Ratio***: Low value of ***0.54***. This confirms that as spending increases, there are more teachers to accommodate students, so the student-to-teacher ratio decreases. This confirms the negative correlation shown in Figure 10.

Significance of Variables

Principal Component Analysis is an ***unsupervised*** learning method that can take high-dimensional data (many columns), and reduce them to a small number of variables while still retaining the maximum amount of data (Jaadi). Previously, our results comprised analyses on a small number of selected variables from the dataset. PCA allows us to look at the dataset ***as a whole*** and determine which socioeconomic variables are ***most significant*** when determining educational outcomes.

The goal of PCA is to find the properties of a line that maximizes the variance of each of the variables being analyzed. This variance is equivalent to the distances between each projected point on the line and the origin.

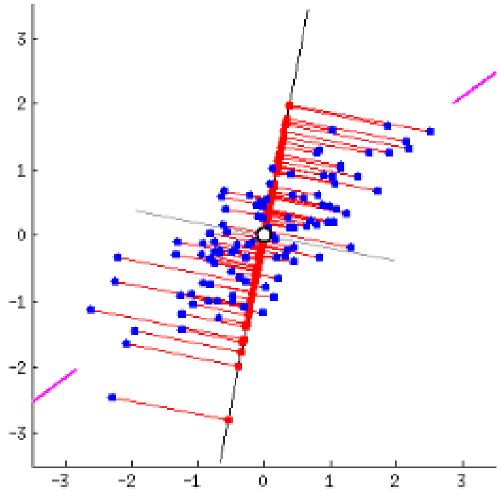


Figure 20. Graph showing variances for a dataset with 2 fields. (Credit: Zakaria Jaadi)

Then, a matrix is created containing the covariances for each pairing of variables. For a dataset with three variables, the matrix would resemble the following:

$$\begin{bmatrix} Cov(x,x) & Cov(x,y) & Cov(x,z) \\ Cov(y,x) & Cov(y,y) & Cov(y,z) \\ Cov(z,x) & Cov(z,y) & Cov(z,z) \end{bmatrix}$$

Figure 21. Covariance Matrix

$$Cov(a, b) = \frac{\sum_{i=1}^{n} (a_i - \bar{a})(b_i - \bar{b})}{N-1}$$

Figure 22. Covariance Function Formula

The ***eigenvectors*** of this matrix correspond to the ***linear combinations*** of each principal component. Each of the elements in the vectors corresponds to a "loading score" of each variable. This ***loading score*** essentially represents the importance of the variable within the principal component. Each of the ***eigenvalues*** of the matrix corresponds to how much of the variance is accounted for by each component.

In our use of PCA, we utilized 14 socioeconomic variables and attempted to reduce this to the optimal number of components. A scree plot, showing the proportion of variance accounted for by each component was generated.

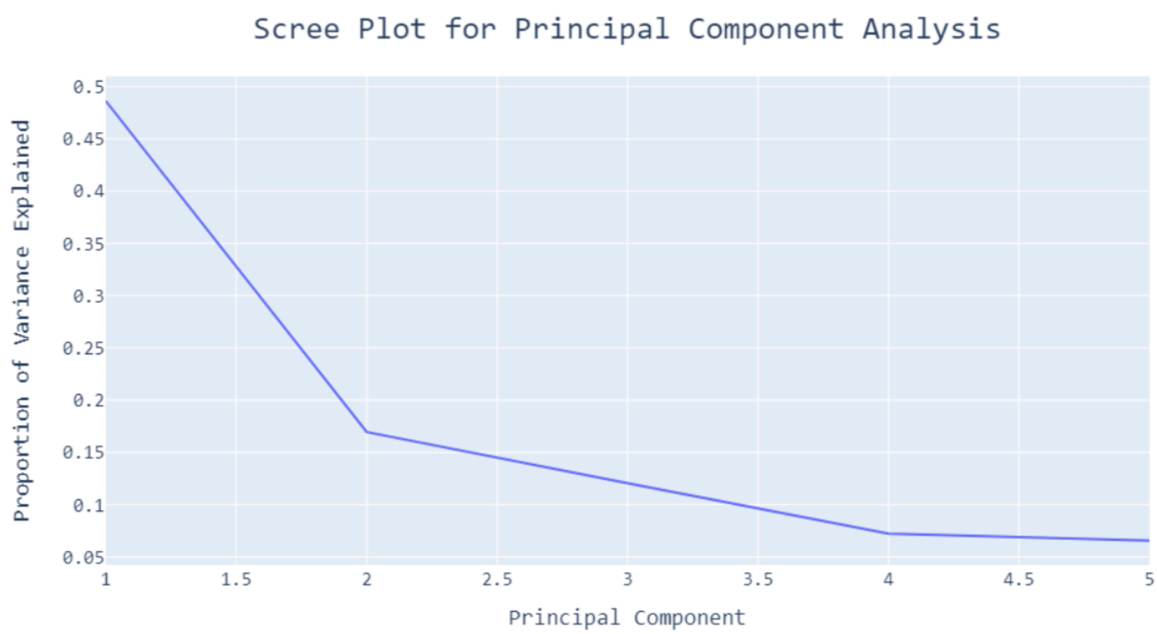


Figure 23. Scree Plot showing Proportion of Variance for each number of PCs

After taking the sum of the first four principal components, a sum of 0.85 is attained. This is well over ***80%*** explained variance and is a number of components that will suffice according to prior literature (Toth).

Finally, we mapped the matrix of loading weights to a heatmap. For each pairing of socioeconomic variables and principal components, a certain value is given. This value essentially describes the "***importance***" of the variable within that principal component. The principal components are also ordered by importance, where PC1 accounts for more variance than PC2, and so on.

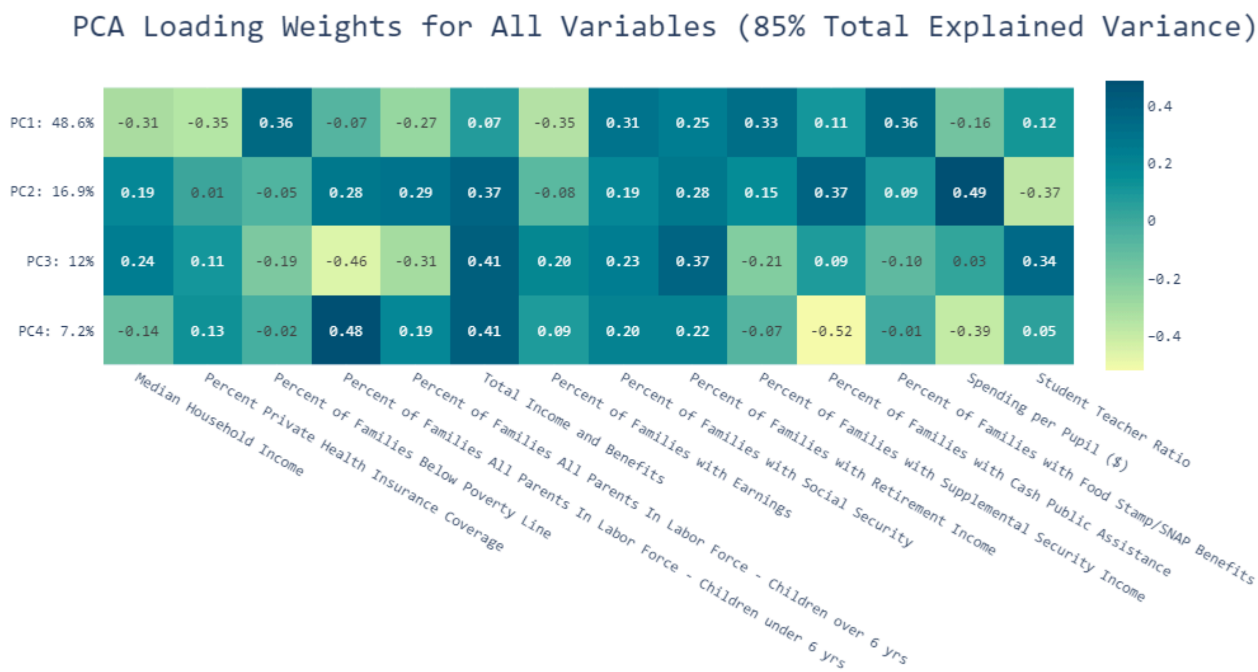


Figure 24. Heatmap of loading weights for all variables for each principal component.

The heatmap shows that for PC1, the highest valued variables are:

- ***Percent of Families below the Poverty Line*** *(+0.36)*
- ***Percent of Families with Food Stamp Benefits*** *(+0.36)*
- ***Percent of Families with Supplemental Security Income*** *(+0.33)*
- ***Percent of Families with Social Security Income*** *(+0.31)*

According to the analysis, these variables hold a position of importance within the dataset as a whole, and most likely have the ***strongest positive influence*** on educational outcomes.

The lowest valued variables can indicate which variables have the strongest ***negatively correlated*** influence on the data as a whole.

In PC1, these variables were:

- ***Percent of Private Health Insurance Coverage Percent of Families with Earnings*** *(-0.35)*
- ***Median Household Income*** *(-0.31)*
- ***Percent of Families with both Parents in the Labor Force [with children over 6 years of age]*** *(-0.27)*

This seems logical as these four variables with negative values are all inversely related to those with positive values. For example, a ***higher*** percentage of poverty would indicate ***lower*** educational performance on average, while a higher percentage of families with health insurance would indicate higher educational performance on average.

In PC2, we can see some of the less important variables such as:

- ***Spending per Pupil*** *(+0.49)*
- ***Total Family Income and Benefits*** *(+0.37)*
- ***Percent of families with cash public assistance*** *(+0.37)*

While these factors are not as influential in determining outcomes as those prevalent in PC1, these are still much more important than those in PC4 such as the ***Percent of Families with both parents in the Labor Force (Children under 6 years of age) (+0.48)***. Essentially, the Principal Component Analysis model provides a way to see which socioeconomic factors are the most ***influential***.

Conclusions

Poverty in the United States

The analyses on household income demographics and poverty rates demonstrate that poverty is a ***prevalent*** issue across the nation. Furthermore, a significant proportion of the country's population was found to be ***dependent*** on government insurance or ***uninsured***. Some notable findings are:

- Approximately ***1 in every 6*** families with school-going children are below the poverty line.
- The average of the Median Household Incomes of all school districts came to ***$76,471***. Income is generally skewed lower, while there are some outlier districts with extremely high income.
- States in the New England region such as New Hampshire, Connecticut, Massachusetts, and New Jersey have extremely ***low*** poverty rates.
- States in the North such as North Dakota, Wisconsin, Iowa, and Minnesota have ***low*** poverty rates.
- States in the deep south such as Mississippi, Alabama, Georgia, Kentucky, and Louisiana show ***extremely high*** poverty rates.
- Approximately 4.58% of families are ***uninsured***.
- Approximately 35.4% of families are covered by ***publicly funded health insurance***.
- States on the Pacific Coast and the Eastern half of the U.S. have quite ***low*** rates of uninsurance.
- States in the Mountain West as well as Alaska show quite ***high*** rates of uninsurance.

Government Spending on Education by State

Spending was analyzed with respect to geographic location as well as political alignment. Southern and western states were found to ***spend less*** per pupil, while northeastern states were found to spend more. Through the use of linear regression, it was concluded that conservative states generally spend less per student than liberal states, with a correlation of determination of ***r^2 = 0.42***.

Effect on ACT Scores

Through the use of local regression, ACT scores were found to ***increase*** as spending per student ***increased***. This increase is ***sharper*** in the lower rates of spending and then plateaus with higher rates of spending.

ACT scores were found to be relatively ***stagnant*** when compared to poverty rates, but after poverty rates exceed 17%, states seem to show a ***decline*** in standardized test performance.

Advanced Course Enrollment

K-Means clustering was used to model the relationship between enrollment in advanced mathematics courses, student-to-teacher ratio, and government spending per student. Some of the findings include:

- As student-to-teacher ratios ***increase***, spending per student ***decreases***.
- No states show high student-to-teacher ratios and high advanced mathematics enrollment ***simultaneously***.
- The majority of states fall within the cluster which shows ***moderate*** enrollment in advanced mathematics with a ***relatively low*** student-to-teacher ratio.
- California and Utah show the ***highest*** average student-to-teacher ratios.
- Maryland and Virginia show the ***highest*** average enrollment in advanced mathematics courses.

Confirmation of Correlations

The Cosine Similarity test ***confirmed*** the correlations between various variables that were tested in prior analyses. Notable values are:

- Median Household Income & Spending per Student *(0.90)*
- ACT Composite & Spending per Student *(0.89)*
- Enrollment in Advanced Mathematics & Median Household Income *(0.83)*

- Spending per Student & Student-to-Teacher Ratio (0.54)

Significance of Variables

The Principal Component Analysis model showed which socioeconomic factors were ***most important*** within the dataset and had the most influence on educational outcomes. This was done by viewing the loading scores for each of the variables within each of the principal components. Seven of the highest were:

- *Percent of Families below the Poverty Line (+0.36)*
- *Percent of Families with Food Stamp Benefits (+0.36)*
- *Percent of Families with Supplemental Security Income (+0.33)*
- *Percent of Families with Social Security Income (+0.31)*

- *Percent of Private Health Insurance Coverage Percent of Families with Earnings (-0.35)*
- *Median Household Income (-0.31)*
- *Percent of Families with both Parents in the Labor Force [with children over 6 years of age] (-0.27)*

Next Steps

The results clearly show that ***greater funding*** for students impacts them ***positively***, and ***less poverty*** also correlates to ***better*** student ***performance***. These circumstances can be addressed in a multitude of ways. Specifically, addressing the variables outlined by the primary and secondary components in the ***PCA Analysis*** should, in theory, lead to the largest change in educational outcomes.

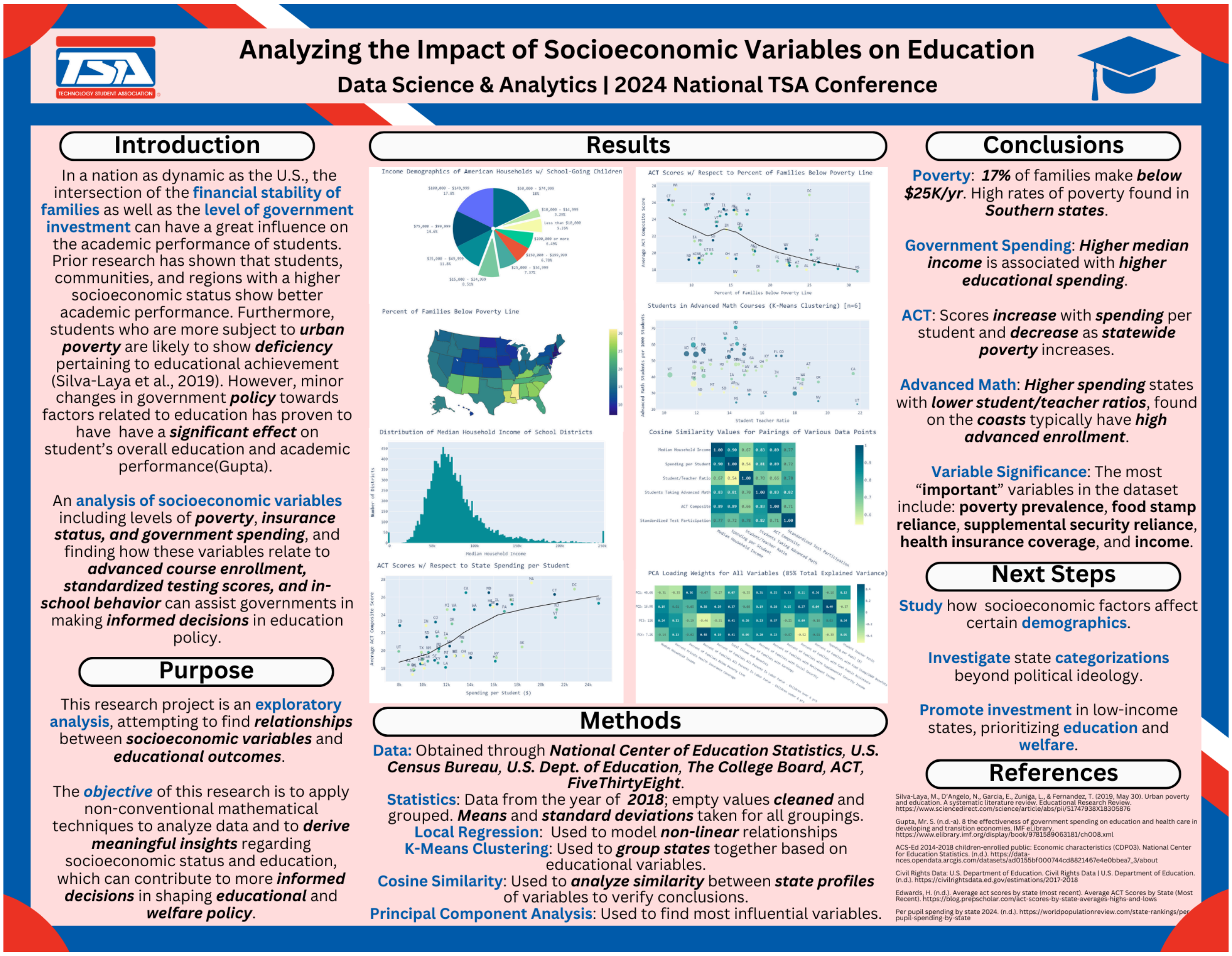
Further research pertaining to the disparity in socioeconomic status among various geographic regions, as outlined by the ***K-Means Clustering Map***, will be beneficial. The use of causal inference techniques to analyze the disparities between certain regions can further increase our understanding of how to make the education system more effective.

When considering funding, governments may either choose to invest more into the ***public*** schooling system, or another viable approach would be to encourage enrollment into ***privately*** funded schools by providing dividends to families. Both of these solutions seem viable in theory, as they both have the potential to result in greater funding for students. However, further research is necessary to determine which of these approaches is more beneficial.

Concerning the issue of poverty, there are once again two major solutions to the issue. One solution is ***greater government spending*** on social programs for healthcare, food stamps, and supplemental income. Another solution is ***reducing the tax burden*** on individuals and increasing tax breaks given by the government.

Regardless of which solutions are implemented, it is crucial to understand the ***necessity of adequate financial resources in education***. When students have less economic burden, they can devote more attention to their education. As shown by the state-by-state analyses, these trends of poverty and economic burden tend to be focused on ***specific geographic areas***. The nation as a whole needs to be ***attentive*** towards these areas to ***break the cycle*** of poverty and ***improve*** educational ***outcomes***.

Digital Scientific Poster



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Appendix

Jupyter Notebook Python Code Files (.ipynb):

https://drive.google.com/drive/folders/1TtmWLzW2cxz1VE4hygj-On67rXmvp_CD?usp=sharing

Spreadsheets from Dataset (.xlsx):

https://drive.google.com/drive/folders/1D8z-mnpB8RVkKulTN9H6ggVGsSbskqot?usp=sharing

Cleaned Spreadsheets (.csv):

https://drive.google.com/drive/folders/1veEljVB8_AlFsJOIiGLnO-FPc4zPH-H5?usp=sharin

Work Logs (.pdf):

https://drive.google.com/drive/folders/1xfFkscAgzaV9tNRKwvAqGzv3p3JgPF_L?usp=sharing

Citation of Font

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Copyright Checklist & Release Forms

STUDENT COPYRIGHT CHECKLIST
(for students to complete and advisors to verify)

1) **Does your solution to the competitive event integrate any music?** YES NO

If NO, go to question 2.

If YES, is the music copyrighted? YES NO

If YES, move to question 1A. If NO, move to question 1B.

1A) Have you asked for author permission to use the music in your solution and included that permission (letter/form) in your documentation? If YES, move to question 2. If NO, ask for permission (OR use royalty free/your own original music) and if permission is granted, include the permission in your documentation.

1B) Is the music royalty free, or did you create the music yourself? If YES, cite the royalty free music OR your original music properly in your documentation.

CHAPTER ADVISOR: Sign below if your student has integrated any music into his/her competitive event solution.

I, ______________________ (chapter advisor), have checked my student's solution and confirm that the use of music is done so with proper permission and is cited correctly in the student's documentation.

2) Does your solution to the competitive event integrate any graphics? YES NO

If NO, go to question 3.

If YES, is the graphic copyrighted, registered and/or trademarked? YES NO

If YES, move to question 2A. If NO, move to question 2B.

2A) Have you asked for author permission to use the graphic in your solution and included that permission (letter/form) in your documentation? If YES, move to question 3. If NO, ask for permission (OR use royalty free/your own original graphic) and if permission is granted, include the permission in your documentation.

2B) Is the graphic royalty free, or did you create your own graphic? If YES, cite the royalty free graphic OR your own original graphic properly in your documentation.

CHAPTER ADVISOR: Sign below if your student has integrated any graphics into his/her competitive event solution.

I, ______________________ (chapter advisor), have checked my student's solution and confirm that the use of graphics is done so with proper permission and is cited correctly in the student's documentation.

3) Does your solution to the competitive event use another's thoughts or research? YES NO

If NO, this is the end of the checklist.

If YES, have you properly cited other's thoughts or research in your documentation? If YES, this is the end of the checklist.

If NO, properly cite the thoughts/research of others in your documentation.

CHAPTER ADVISOR: Sign below if your student has integrated any thoughts/research of others into his/her competitive event solution.

I, ______________________ (chapter advisor), have checked my student's solution and confirm that the use of the thoughts/research of others is done so with proper permission and is cited correctly in the student's documentation.

Release forms:

https://drive.google.com/drive/folders/1ML2VZCEaDzHLalB2godNOgaYIBnuHwSB?usp=sharing